

JONEIL CAOILE

joneilcaoile@gmail.com | (424) 421-9173 | Torrance, CA 90502 | [linkedin.com/in/joneilcaoile](https://www.linkedin.com/in/joneilcaoile) | [joneilcaoile.github.io](https://github.com/joneilcaoile)

PROFESSIONAL SUMMARY

M.S. Biomedical Engineering candidate (Stevens, 3.9 GPA) with a B.S. in Kinesiology and 400+ hours of clinical experience. Bridges engineering rigor and clinical workflow understanding for customer-facing technical roles. Background in human-centered design, simulated-use workflow design, and translating technical outputs into clinician-actionable insights.

TECHNICAL SKILLS

Applications & Communication: Human-centered design, usability testing, simulated-use workflow design, clinical workflow analysis, technical communication, data visualization, customer requirements analysis, training material development.

Clinical & Analytical: EMG signal analysis, motion capture (OptiTrack), biomechanics, gait & balance analysis, cardiopulmonary physiology, clinical research methodology, IRB/HIPAA exposure (coursework).

Technical: Python (NumPy, SciPy, Matplotlib, Streamlit), MATLAB, R, SQL (SQLite, PostgreSQL), Excel, Qualtrics, Git, SolidWorks; Figma for clinical workflow mockups.

EDUCATION

Stevens Institute of Technology, *M.S. Biomedical Engineering* Expected May 2026 | GPA 3.9
Concentration: Medical Devices & Clinical Development | Provost Master's Scholarship
Relevant Coursework: Machine Learning, Clinical Research and Design, Biofeedback, Physiology, BMI, Biostatistics
California State University, Long Beach, *B.S. Kinesiology: Exercise Science* Dec 2020 | Cum Laude

PROJECTS

Chest Drain Air Leak Monitor – Inline Retrofit Cartridge (Advisor: Dr. Jinho Kim) Jan 2026 – May 2026 (Expected)
• Designed simulated-use workflows for an inline retrofit cartridge that adds digital air leak monitoring to existing chest drains, defining clinician interaction with the sensor outputs.
• Developed user-facing display concepts and alarm logic on the cartridge LCD, grounded in clinical scenarios and Cerfolio-sourced threshold logic.

Parkinson's Tremor-Suppression System (Product & Prototyping Lead) Sep 2025 – Dec 2025
• Designed a wearable EMG-based cuff concept for Parkinson's patients, focusing on wearability, comfort, and real-world use by individuals with motor impairments.
• Built a Python (NumPy, SciPy) simulation pipeline (4-8 Hz Butterworth bandpass, RMS envelope) and communicated design tradeoffs to a multidisciplinary team grounded in measurable physiological outcomes.

Clinician-Facing Reliability Tool: Fall-Risk Assessment (Advisor: Dr. Antonia Zaferiou) Jan 2025 – May 2025
• Built a SQLite-backed validation tool that cleaned a 77-patient clinical survey dataset and computed inter-rater reliability metrics (Krippendorff's Alpha, percent agreement).
• Designed a clinician-facing decision-support dashboard, iterated outputs with faculty stakeholders, and translated technical reliability scores into language non-technical clinicians could act on.

EXPERIENCE

Graduate Researcher Jan 2025 – Present
Stevens Institute of Technology
• Design test protocols and instrumentation for a retrofit chest drain sensor module capstone in Dr. Jinho Kim's lab (Jan 2026–Present).
• Previously: Dr. Antonia Zaferiou's Musculoskeletal Control and Dynamics Lab (Jan-May 2025); see project work above.

Physical Therapy Aide 2016 – 2023 | 400+ Hours
Various Physical Therapy Clinics (Orthopedic, Pediatric, Geriatric)
• Assisted therapists with patient sessions, device setup, and treatment workflow across three clinical settings. Built first-hand understanding of end-user needs for medical device design and testing.

CERTIFICATIONS

Microsoft / LinkedIn Career Essentials (Apr 2026): Data Analysis, Cybersecurity (95%), Business Analysis (95%), Project Management; Atlassian Agile Project Management (94%); Aha! Product Management Certified.